

NIVEDITHA SRIKANTH

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EXPERIENCE

The Cigna Group

Bengaluru, KA, India

Data Science Associate Analyst

Jan 2026 – Apr 2026

- Developed a composite risk-scoring framework using log-scaled transformations and percentile-based normalization (CUME_DIST) for healthcare claims data, improving detection coverage by **20%** and contributing to **\$400K–\$700K** in potential savings
- Assisted in applying BIRCH clustering and Local Outlier Factor (LOF) to segment healthcare providers, identifying **18%** high-risk providers
- Supported feature engineering (**40+ features**) across **10M+** records and AWS deployment of scoring pipelines (S3, Athena, SageMaker)

PROJECTS

Healthcare Claims Fraud & Anomaly Detection | Python, Scikit-learn, PostgreSQL, SQL, pandas, psycopg2, Matplotlib, Power BI

- Developed an ETL workflow on **1M+ Medicare claims**, detecting **3%** anomalies using rule-based SQL flagging and provider outlier scoring (BIRCH, LOF), and prioritizing the top **1%** high-risk claims (**128K**) through an interactive Power BI dashboard

Mclust-Based Gene Expression Clustering for Immune Checkpoint Analysis | R, Mclust, PCA, t-SNE, UMAP, ggplot2, KEGG

- Clustered **41K+ mouse aortic endothelial genes** (GEO: GSE57329) via PCA + Gaussian Mixture Models to map immune check points in T2D-atherosclerosis; identified hub genes **Cd4, Cxcl10, Fmo3** via STRING/CytoHubba MCC ranking, validated with silhouette avg. **0.81**, t-SNE, and KEGG/Reactome enrichment

Cerebrovascular Accident Prognosis using ML | Python, scikit-learn, XGBoost, Random Forest, Decision Tree, AUC-ROC

- Developed and evaluated supervised ML models on **43,400** patient records of clinical data (age, BP, cholesterol, history), achieving **99.9% accuracy** with Random Forest and **97.9%** with XGBoost for predicting cerebrovascular disease risk (WCONF 2023 – Best Presentation Award)

Scientific Paper Clustering | Python, scikit-learn, TF-IDF, PCA, t-SNE, UMAP, K-Means, NLP, topic modelling, Jupyter Notebook

- Built a scientific paper clustering pipeline using NLP embeddings, applying PCA and K-Means on **10K+** arXiv research papers with multi-language preprocessing and TF-IDF variants, improving literature retrieval efficiency by **40%**

TECHNICAL SKILLS

- ML & AI:** Python, R, scikit-learn, PyTorch, TensorFlow, XGBoost, Random Forest, BIRCH, LOF, GMM (Mclust), KMeans, PCA, t-SNE, NLP (TF-IDF, BERT, spaCy), CNNs, Sequence Models, Feature Engineering, Anomaly Detection, Model Evaluation
- Data & Cloud Infrastructure:** SQL (Athena, PostgreSQL, MySQL), AWS (S3, Glue, SageMaker, Lambda, Redshift), ETL Pipelines, Docker, Git, Power BI, Tableau, Jupyter, MS Excel

EDUCATION

Rajalakshmi Engineering College, Anna University

Chennai, TN, India

Bachelor of Technology, Biotechnology

Oct 2021 – May 2025

- Coursework:** Python for Machine Learning, Probability & Statistics, Data Structures & Algorithms, Bioinformatics
- Concurrently completed a **16-month Micro-Degree Credit-Linked Program in Computer Science**, IIT Guwahati (Daksh Gurukul): Data Structures & Algorithms, Discrete Mathematics, Database Systems, Cloud Architecture

ACHIEVEMENTS & CERTIFICATIONS

- Competitions:** Top 10 Performer at Databricks Spark-Wars 2023 (data engineering challenges); Top 20 Teams at Shell.ai Hackathon 2023 (AI/data solutions for sustainable, affordable energy)
- Awards:** Best Presentation Award at IEEE WCONF 2023 for Cerebrovascular Accident Prognosis, 97–99% predictive accuracy
- Recognition:** IEEE Reviewer for Scopus-indexed conferences, 8 conferences (2023–2024)
- Certifications:** IBM Professional Data Science Certificate; Google Data Analytics Professional Certificate

PUBLICATIONS

Brain Tumor Segmentation using Multi-Task Learning (GlobalNet, FusionNet) | IEEE CSNT, 2023 (Best Presentation and Paper Award)

- 3-architecture deep learning pipeline (**GlobalNet, Multi-task Learning, FusionNet**) for brain tumor segmentation; improved accuracy and resilience across multi-modal medical imaging

Malware Classification using ML with Feature Selection (Cuckoo Sandbox) | Procedia Computer Science, 2024

- KNN + Boruta automated feature selection for malware family prediction; **93%** precision, **96%** recall, **92%** F1 on phishing detection in Cuckoo sandbox environment

Speech Signal Segmentation using Kernelized Deep Networks and Clustering | IEEE CSNT, 2023

- Kernelized deep networks with **MFCC/LPC + spectral clustering (graph Laplacian)**; superior segmentation accuracy with reduced complexity on TIMIT and BURN benchmark corpora